KUMBH CARE

SMART HEALTH EMERGENCY DETECTION & RAPID RESPONSE FOR KUMBH MELA

Problem: Delayed identification and response to health emergencies in crowded Kumbh Mela zones.

THEME	TEAM NAME	TEAM ID	STATE / UT
Healthcare / Smart Healthcare	Add registered team name	Add registered team ID	Uttar Pradesh

DETECT EARLY • RESPOND QUICKLY • SAVE LIVES

Problem – Proposed Solution

A practical coordination platform; proposed features require prototype validation.

THE PROBLEM

- Dense crowds make patients difficult to locate.
- Emergencies may go unnoticed or be reported late.
- Visitors may not know how to share their exact zone.
- Medical teams need clear incident details and assignment status.

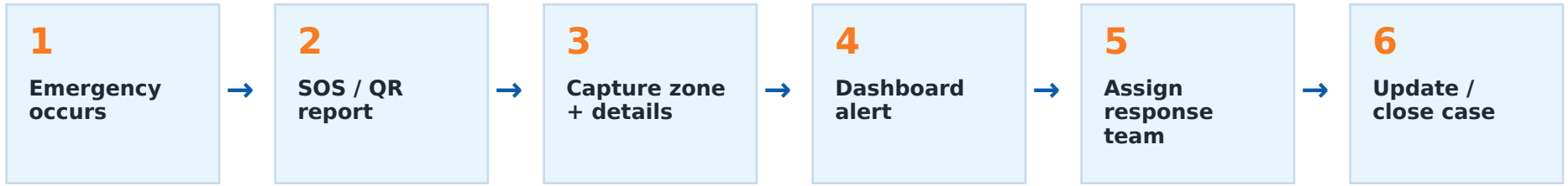
KUMBH CARE SOLUTION

1. One-tap SOS or volunteer-assisted report.
2. QR codes at approved help points for zone-based requests.
3. Live incident map with severity and status.
4. Alerts to an authorized medical coordinator for team assignment.
5. Track each case from report to closure.

UNIQUE VALUE **Combines location-aware reporting, incident visibility and response tracking in one workflow.**

Technical Approach

Prototype workflow: report → locate → notify → assign → respond → close.



SUGGESTED TECHNOLOGY STACK

Mobile: Flutter | Dashboard: React.js | Backend: Python / FastAPI
Database: Firebase or PostgreSQL | Map: OpenStreetMap | Notifications: Firebase Cloud Messaging
QR reporting: QR-code library | Optional AI: classify incoming reports for coordinator review

FALLBACKS & SAFETY

Manual zone selection if GPS fails • Queue reports if connectivity drops • Volunteer-assisted reporting • Human review for severity and assignment • Existing emergency services remain primary.

Repository / Demo: Add GitHub or GitLab URL and demo video when available.

AI tools used: disclose tools actually used; do not claim unimplemented AI features.

Impact Potential

Designed to improve information flow and coordination—not replace emergency services.

TARGET USERS

Pilgrims • elderly visitors • people with disabilities • volunteers

HEALTHCARE

Clearer incident details; visible status; better triage support

OPERATIONS

Map reported incidents; support assignment and resource planning

STAKEHOLDERS

Kumbh administration • health departments • hospitals • ambulance teams • police • volunteers

MEASURE SUCCESS IN A PILOT

Alert delivery time • time to team assignment • successful alert delivery rate • location accuracy • incident closure rate.

Feasibility and Risk Analysis

Start with a controlled prototype; deploy in a live venue only with official approval.

RISK / LIMITATION	MITIGATION	FEASIBILITY NOTE
Network congestion	Queue reports; define alternate contact procedures	Prototype can simulate offline cases
GPS errors / dense buildings	Manual zone selection and landmarks	Test location accuracy at known points
False or duplicate alerts	Verification, deduplication and human review	Use test cases before field use
Privacy of medical details	Collect minimum data; role-based access	Apply secure storage and retention rules
Unassigned / overload cases	Priority queue and escalation timer	Requires trained coordinator and staffing

Overall: Technically feasible as a prototype; live deployment needs health authority coordination, training, privacy safeguards and reliable escalation procedures.

Team

Replace placeholders with your actual registered team members and roles.

TEAM LEAD

Member: [Add name]
Planning, coordination, pitch

APP DEVELOPER

Member: [Add name]
SOS screen and user flow

BACKEND DEVELOPER

Member: [Add name]
Database and alert processing

MAPPING LEAD

Member: [Add name]
Zone map and location checks

TESTING / RESEARCH

Member: [Add name]
User testing and documentation

CHALLENGES

Designing a simple interface under stress • handling inaccurate GPS and weak networks • preventing false alerts • coordinating with real medical responders.

OUR MISSION: Make emergency assistance more accessible, organized and responsive.

References

Use sources actually consulted; verify current official guidance before final submission.

WHO — Mass gatherings

https://www.who.int/health-topics/mass-gatherings

WHO — Emergencies

https://www.who.int/health-topics/emergencies

National Disaster Management Authority, India

https://ndma.gov.in/

Ministry of Health and Family Welfare, India

https://mohfw.gov.in/

OpenStreetMap — Mapping data

https://www.openstreetmap.org/

Firebase documentation

https://firebase.google.com/docs

Flutter documentation

https://docs.flutter.dev/

Prototype status: proposed concept. Add your repository, demo link and any research papers or official Kumbh-specific guidelines used.